

# Underwater ROV

## Programming Curriculum

A hands-on Python curriculum for students building and programming an underwater ROV (Remotely Operated Vehicle) controlled by a Raspberry Pi running Ubuntu.

Each topic lives in its own folder with a `lesson.md` (the lesson content) and any example code that goes with it.

## Hardware assumed throughout this course

- Raspberry Pi running **Ubuntu** (not Raspberry Pi OS)
- USB webcam plugged into the Pi
- Three BlueRobotics-style brushless thrusters (left, right, vertical), each with its own ESC, wired to a **SparkFun Pi Servo Hat** (PCA9685-based) plugged onto the Pi's GPIO header
- A topside laptop on the same network as the Pi (connected via tether/switch or Wi-Fi), used to view the

camera stream and send control commands

- A USB/Bluetooth game controller plugged into the laptop, for piloting
- **PyCharm Professional** installed on the topside laptop for writing, running, and debugging code — including code that runs on the Pi over SSH

## Foundations

Skills you need before writing ROV code. Work through these first if you're new to Linux or version control.

| Lesson                                   | Folder     |
|------------------------------------------|------------|
| Linux Shell Basics                       | shell/     |
| Git Basics                               | git/       |
| PyCharm for Remote ROV Development       | pycharm/   |
| Debugging: print() vs PyCharm's Debugger | debugging/ |

## ROV Lessons

| # | Lesson                               | Folder     |
|---|--------------------------------------|------------|
| 1 | Streaming Video from a USB Camera    | camera/    |
| 2 | Thruster Control                     | thrusters/ |
| 3 | Networked Control (Piloting the ROV) | control/   |